

frequency source and enough counting stages (e.g. 25) to measure the frequency accurately.

Applications for this high speed counter include the high speed A/D converter proposed by Hurrell and Silver, [1]. Since it can count Josephson oscillations directly, the counter could also be used to make a very high accuracy A/D converter based directly on the Josephson voltage frequency relation. It is speculated that it may also be useful in very wide dynamic range photon counting.

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Model for Modulation Doped Field Effect Transistor

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Abstract—A model describing C - V and I - V characteristics of modulation doped FET's is proposed. The model takes into account the change in the Fermi energy with the gate voltage. At high two dimensional electron concentrations, the equations of the model for the charge control by the gate voltage become similar to the equations of the charge control model where the thickness d of AlGaAs layer should be substituted by $d + \Delta d$ and Δd is the effective width of the potential well ($\approx 80 \text{ \AA}$). Another important prediction of the model is the existence of the "subthreshold" current. A very good quantitative agreement is obtained with our experimental I - V curves using the measured values of the low field mobility and the source resistance.

MUCH progress has been made in the fabrication of modulation doped field effect transistors suitable for practical applications [1-3]. Ring oscillators operated with a propagation delay of 17.1 ps at 77°K comparable to the performance of Josephson devices [4]. A model describing the C - V and I - V characteristics of the modulation doped structures has been developed [5]. In this model, the amount of charge transfer across the interface is found by equating the charge depleted from the (Al/Ga)As to the charge accumulated in the potential well. A solution is then found such that the Fermi level is constant across the hetero-interface. The charge depleted from the (Al/Ga)As is given by

$$n_s = \sqrt{\frac{2eN_d}{q}(\Delta E_c - E_{F2} - E_{Fi}) + N_d^2 d_i^2 - N_d d_i} \quad (1)$$

where E_{F2} is the separation of the conduction band edge

and the Fermi level in the bulk (Al/Ga)As, N_d is the doping level in (Al/Ga)As layer, ϵ is the dielectric permittivity of (Al/Ga)As, ΔE_c is the discontinuity in the conduction band at the interface, E_{Fi} is the Fermi level at the interface relative to the GaAs conduction band, edge, and d_i is the thickness of the intrinsic (Al/Ga)As layer. The charge accumulated in the potential well is given by

$$n_s = \frac{DkT}{q} \ln [(1 + e^{q/kT(E_{Fi} - E_0)}) \cdot (1 + e^{q/kT(E_{Fi} - E_1)})] \quad (2)$$

where $E_0 = \gamma_0 n_s^{2/3}$ and $E_1 = \gamma_1 n_s^{2/3}$ are the positions of the first two allowed energy levels in the triangular well. The energy reference in the well is the GaAs conduction band edge at the heterojunction. The constants γ_0 , γ_1 and D ($= qm^*/\pi\hbar^2$, density of states) are derived in the formalism of the triangular well but have been adjusted slightly to obtain closer agreement with measured subband splitting and electron effective mass. Placing a Schottky gate on the (Al/Ga)As results in a certain amount of depletion beneath the gate. If the (Al/Ga)As layer is thin enough, or a sufficiently large negative gate voltage is applied, the gate depletion and junction depletion regions will overlap, in which case (1) must be replaced by

$$n_s = \frac{\epsilon}{qd} [V_g - (\phi_b V_{P2} + E_{Fi} - \Delta E_c)] \quad (3)$$

where ϕ_b is the Schottky barrier height, V_g is the gate voltage, and $V_{P2} = qN_d d_d^2 / 2\epsilon$, d_d being the thickness of the doped (Al/Ga)As beneath the gate and $d = d_d + d_i$. The

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simultaneous solution of (2) and (3) then yields the electron concentration n_s in the potential well for cases with n_s greater than zero and less than the equilibrium n_s found from the simulations solution of (1) and (2).

If the (Al/Ga)As is too thick, or a sufficiently large positive gate voltage is applied, a parallel conduction path in the (Al/Ga)As is created. For a given d_d the maximum gate threshold voltage which affects the two-dimensional electron gas can be obtained by equating (1) and (3) using the Fermi energy at the interface in equilibrium. This condition implies that the (Al/Ga)As underneath the gate is fully depleted but the gate and junction depletion depths do not interpenetrate.

In the model [5], the total charge in the two-dimensional electron gas is found neglecting the variation of the Fermi level with the gate voltage

$$n_s = \frac{\epsilon}{qd} (V_g - V_{off}^0) \quad (4)$$

where

$$V_{off}^0 = \phi_b - \Delta E_c - V_{P2}$$

is the threshold voltage. Using (4), I - V characteristics of a modulation-doped FET may be then found assuming that current saturation is reached when the electric field at the drain side of the gate is equal to the saturation field $F_s = v_s/\mu$ [5, 6], where v_s is the saturation velocity and μ is the low field mobility. However, in a typical normally-off transistor, E_{Fi} can be larger than 0.15 V when the device is fully on, causing the charge to be significantly overestimated. The Fermi level, E_{Fi} , determines the width of the potential well at the interface. As the Fermi level drops, the well becomes wider and the spatial distribution of the electrons changes. As shown below, this effect leads to a substantial increase in the effective distance from the two-dimensional electron gas to the gate, to a small change in the threshold voltage, to a very pronounced decrease of the transconductance near the threshold and to a "subthreshold" current.

Equation (2) is a quadratic equation with respect to $\exp(qE_{Fi}/kT)$. The solutions of (2) at 300°K, 77°K and 4°K are shown in Fig. 1 (solid lines). The calculated density n_s for 300°K is about half of the density predicted by a less accurate three-dimensional gas model which neglects the quantization in the potential well and uses the Joyce-Dixon approximation [8]. For the values of n_s between $5 \times 10^{11} \text{ cm}^{-2}$ and $1.5 \times 10^{12} \text{ cm}^{-2}$, these dependencies can be approximated as (shown in dashed lines)

$$E_{Fi} = \Delta E_{Fo}(T) + a \cdot n_s \quad (5)$$

where $a \cong .125 \times 10^{-16} \text{ V} \cdot \text{m}^{-2}$, and $\Delta E_{Fo} \cong 0$ at 300°K and $\Delta E_{Fo} \cong .025 \text{ V}$ at 77°K and below. Substituting (5) into (3), we find the corrected equation of the charge control model

$$n_s = \frac{\epsilon}{q(d + \Delta d)} (V_g - V_{off}) \quad (6)$$

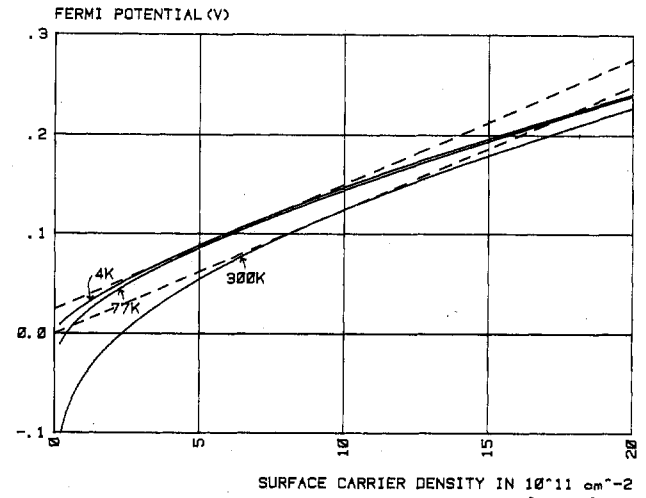


Fig. 1. Fermi potential vs. surface carrier density at 300°K, 77°K, and 4°K (solid lines). Linear approximations (5) are shown as dashed lines.

where

$$V_{off} = V_{off}^0 + \Delta E_{Fo}, \quad (7)$$

and

$$\Delta d = \frac{\epsilon \cdot a}{q} \cong 80(\text{\AA}). \quad (8)$$

Thus Δd gives an important correction especially for a normally off device where d may be of the order of 300 Å or less.

The implications of this model are demonstrated by Fig. 2 where the charge control model of Delagebeaudeuf and Linh (solid line) is compared with (6) (dashed line) and with the exact solutions of (2) and (3) (dotted line). We can see from this figure that if we introduce a threshold voltage V_{off} (see (6)) similar to the threshold voltage of a MOSFET, a substantial "subthreshold" charge exists leading to the subthreshold current. However our modified charge control model given by (6) is quite adequate for the analytical device modelling. The estimation for Δd given by (8) is in a good agreement with the experimental data reported in [8]. The "apparent mobility" for the two-dimensional gas calculated in [8] from the drain conductance vs. gate voltage curve was $55,000 \text{ cm}^2/\text{V} \cdot \text{s}$ at 77°K. The measured value of the Hall mobility was $61,000 \text{ cm}^2/\text{V} \cdot \text{s}$. We estimate that the value of Δd required to explain this difference is about 100 Å in good agreement with (8).

Assuming, as in [5, 6], that the current saturation occurs when the electric field at the drain side of the gate exceeds the velocity saturation field $F_s = v_s/\mu$ and using the Shockley model in order to describe the longitudinal field distribution in the channel below the saturation voltage we find

$$V_{ds}^s = V_g' + V_{sl} - \sqrt{V_g'^2 + V_{sl}^2} + I_{ds}^s(R_s + R_d) \quad (9)$$

and

$$I_{ds}^s = \beta V_{sl}^2 \frac{\sqrt{1 + 2\beta R_s V_g' + V_g'^2/V_{sl}^2} - (1 + \beta R_s V_g')}{1 - \beta^2 R_s^2 V_{sl}^2} \quad (10)$$

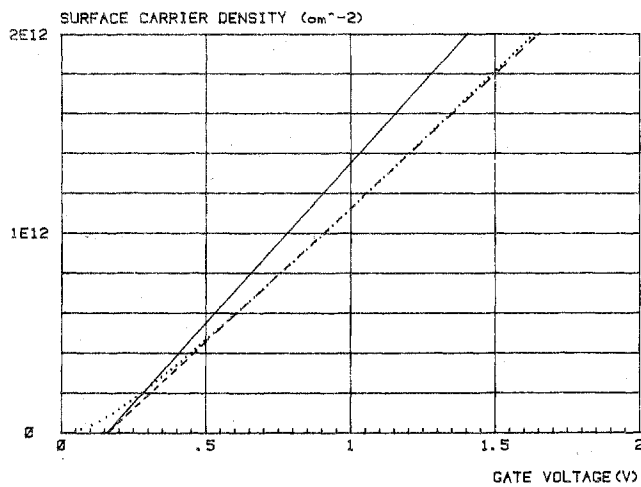


Fig. 2. Surface carrier density vs. voltage difference between gate and channel ($V_{\text{off}}^0 = 0.16$ V and $d = 400$ Å): solid line: simple model proposed in [5], (4) dotted line: numerical solution from (2) and (3) dashed line: present model, (6).

where $V_{sl} = F_s L$, $\beta = \epsilon \mu W / (d + \Delta d) / L$, $V_g' = V_g - V_{\text{off}}$, V_{ds}^s is the drain-to-source saturation voltage and I_{ds}^s is the saturation current.

To test the accuracy of the model, a normally-off modulation doped FET fabricated in our lab was selected and characterized. The low field mobility μ was obtained from the Van der Paw-Hall measurements of the particular wafer from which the FET was fabricated. The gate length is $1 \mu\text{m}$ and the gate is $145 \mu\text{m}$. The heterostructures used to fabricate these FET's were grown by molecular beam epitaxy on CR-doped semi-insulating substrates. The structure consisted of a $1 \mu\text{m}$ undoped GaAs buffer layer, a 100 Å undoped $\text{Al}_x\text{Ga}_{1-x}\text{As}$ layer and a 600 Å n-type $\text{Al}_x\text{Ga}_{1-x}\text{As}$ layer doped with Si to a level of $1 \times 10^{18} \text{cm}^{-3}$. The 600 Å dimension was used to allow fabrication of either normally-on or normally-off devices by recessing the gate. For $x = 0.3$, $N_d = 1 \times 10^{18} \text{cm}^{-3}$ and $d_i = 100$ Å, the doped (Al/Ga)As remaining beneath the gate should be about 300 Å and 450 Å thick to obtain normally-off and normally-on devices respectively.

The calculated and experimental drain saturation currents at room temperature as a function of the gate voltage are shown in Fig. 3. $N_d = 1 \times 10^{18} \text{cm}^{-3}$, $\mu = 0.43 \text{ m}^2/\text{Vs}$ and $R_s = 12 \Omega$ are measured. In this particular device, a value of $v_s = 0.8 \times 10^5 \text{ m/sec}$ was used for best fit for gate voltages below $+0.74$ V. This relatively low value of the saturation velocity may indicate that the effective source resistance of the device is higher than its value measured in the low field region because of the increase of the source resistance at large voltage drop across the source-to-gate section of the device. The result of our extended analysis which takes this effect into account will be published elsewhere. For gate voltages below 0.74 V, the drain current is carried by the 2DEG only and the measured currents fall close to the theoretical curve. Above 0.74 V, the doped (Al/Ga)As layer begins to conduct, which contributes to the drain current. Since the transport in (Al/Ga)As is inferior to that in GaAs, the rate of increase in drain current is reduced. For high

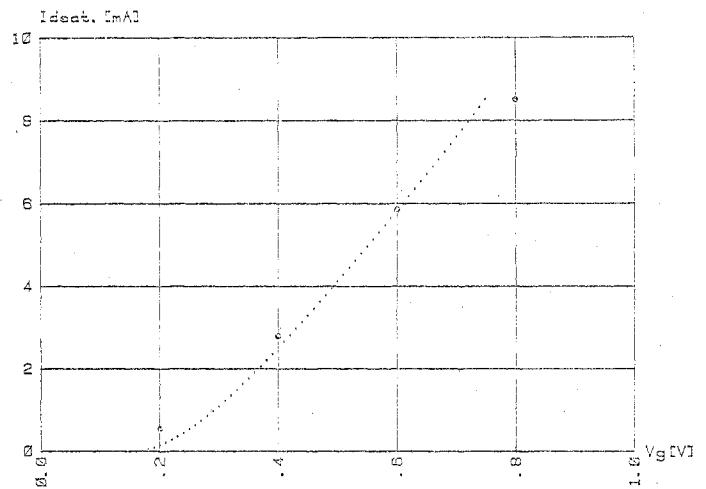


Fig. 3. Saturation current vs. gate voltage for a normally-off device. Measured points are superimposed on the calculated curve.

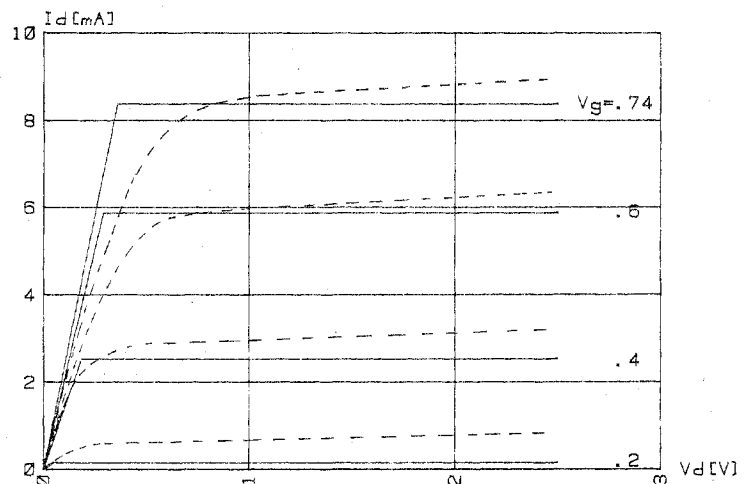


Fig. 4. Source-drain I - V characteristics for a normally-off device—dashed line: measured value; solid line: calculated value.

performance, the gate voltage should be controlled so that the drain current is carried only by the electrons in the GaAs.

Near the threshold voltage, the measured current is larger than that predicted by our analytical model. This effect is due to the subthreshold behavior described above. Decreasing the thickness of the doped (Al/Ga)As would increase the transconductance at the expense of increasing the gate capacitance. It should, however, lead to a better performance since the parasitic capacitances do not change. The I - V characteristics of the device are shown in Fig. 4 which also demonstrates a good fit between the measured results and our model.

In conclusion, it is necessary to account for the dependence of the Fermi energy on gate voltage related to the finite width of the potential well. It is shown that the above consideration leads the effective distance between the gate and the channel to be longer than the actual (Al/Ga)As layer thickness by 80 Å. This leads to a corresponding change of transconductance. Furthermore that dependence leads to subthreshold current. The proposed simple analytical model for FET I - V characteristics shows good agreement with our experimental results.

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A CMOS Structure Using Beam-Recrystallized Polysilicon

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Abstract—A new CMOS structure has been fabricated with the p-channel transistors in a layer of recrystallized polysilicon and the n-channel transistors in adjacent, laterally displaced regions of the underlying single-crystal silicon. The process allows the use of existing circuit layouts with only minor, if any, modification. Moderate quality transistors are fabricated in a layer of recrystallized polysilicon without degrading the characteristics of high-quality transistors simultaneously formed in the substrate. The oscillation period of ring oscillators is close to that calculated, and varies with the transistor dimensions in the expected manner.

INTRODUCTION

MANY recent publications have described the possibility of fabricating moderate-quality MOS transistors with their active channels in a layer of polysilicon recrystallized into a large grain structure with a laser [1], electron beam [2], or other moving heat source [3]. Several novel device structures have advantageously exploited the possibility of forming some devices in the silicon substrate and others in the polysilicon. In particular, novel CMOS structures have been proposed.

In a recently described, stacked-gate CMOS structure [4], one type of transistor is built in the single-crystal substrate, and a transistor of the opposite type is fabricated in a recrystallized polysilicon layer directly above. A common

gate electrode surrounded by gate dielectrics separates the two active channel regions. This structure offers the possibility of area reduction by stacking the transistors on top of each other but complicates the independent optimization of each type of transistor. In addition, the fabrication process is difficult, and existing circuit layouts must be extensively redesigned.

To avoid these limitations in fabricating a CMOS circuit using recrystallized polysilicon, a different approach is reported here. P-channel transistors are fabricated in recrystallized polysilicon and combined with high-performance n-channel transistors in adjacent, but laterally displaced, regions of a single-crystal silicon substrate to form a CMOS integrated circuit. Figure 1 schematically shows the cross section of such a transistor pair. Separating the transistors laterally, rather than stacking them vertically with a common gate, simplifies the processing and allows the use of existing circuit layouts with, at most, only minor modifications.

CMOS integrated circuits are becoming increasingly important for the realization of low-power, high-speed functions. While CMOS is generally easy to use, the possibility of latch-up and the resulting loss of circuit control, may restrict its application. The various techniques that have been used to minimize latch-up may significantly increase the chip size. For example, surrounding the p-channel transistors with oxide layers totally avoids latch-up, but such complete oxide isolation previously required very difficult processing technology

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